

DNA Replication in Prokaryotes and Eukaryotes

A Complete Molecular Biology Teaching and Learning Module

Assessment Package & Answer Key

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Molecular Biology Teaching Package · 2026

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DNA Replication — Multiple Choice Questions

40 Questions with Answers and Explanations

Questions 1-10: Fundamentals

1. The Meselson-Stahl experiment demonstrated that DNA replication is:

- A) Conservative
- B) Semiconservative
- C) Dispersive
- D) Discontinuous

✓ **Answer: B** — After one generation in ^{14}N , all DNA was hybrid density (one ^{15}N strand, one ^{14}N strand), confirming semiconservative replication.

2. DNA polymerase can synthesize DNA in which direction?

- A) $3' \rightarrow 5'$ only
- B) $5' \rightarrow 3'$ only
- C) Both $3' \rightarrow 5'$ and $5' \rightarrow 3'$
- D) Direction depends on the template

✓ **Answer: B** — DNA polymerase catalyzes nucleophilic attack by $3'\text{-OH}$ on the α -phosphate of incoming dNTP. This reaction only works in the $5' \rightarrow 3'$ direction. No known exception exists.

3. Why does DNA replication require a primer?

- A) To provide energy for the reaction
- B) To stabilize the DNA helix
- C) To provide a free $3'\text{-OH}$ group for DNA polymerase
- D) To recruit helicase to the origin

✓ **Answer: C** — DNA polymerase cannot initiate synthesis de novo. It requires a pre-existing $3'\text{-OH}$ group to extend from. Primase synthesizes RNA primers to provide this.

4. In the lagging strand, Okazaki fragments are oriented:

- A) $3' \rightarrow 5'$ relative to fork movement
- B) $5' \rightarrow 3'$ away from the fork
- C) $5' \rightarrow 3'$ toward the fork
- D) Randomly

✓ **Answer: B** — Each Okazaki fragment is synthesized $5' \rightarrow 3'$ (always), but the overall direction of lagging strand synthesis is away from the replication fork.

5. The sliding clamp in *E. coli* is:

- A) A monomer that binds DNA
- B) A homodimer forming a ring
- C) A homotrimer forming a ring
- D) A hexamer that unwinds DNA

✓ **Answer: B** — The β sliding clamp is a homodimer (β_2) forming a toroidal ring. The eukaryotic equivalent (PCNA) is a homotrimer.

6. Which *E. coli* polymerase has 5'→3' exonuclease activity?

- A) Pol I
- B) Pol II
- C) Pol III
- D) All of them

✓ **Answer: A** — Pol I is unique among *E. coli* polymerases in having 5'→3' exonuclease activity, which is essential for removing RNA primers.

7. DNA gyrase is unique because it:

- A) Can only relax positive supercoils
- B) Can introduce negative supercoils using ATP
- C) Is a type I topoisomerase
- D) Is found in both bacteria and humans

✓ **Answer: B** — DNA gyrase is the only topoisomerase that can introduce negative supercoils. It uses ATP and creates a double-strand break to pass another segment through.

8. The end-replication problem affects:

- A) Both linear and circular chromosomes equally
- B) Only circular chromosomes
- C) Only linear chromosomes
- D) Prokaryotic chromosomes only

✓ **Answer: C** — Linear chromosomes cannot fully replicate at their 5' ends because RNA primers at the lagging strand terminus are removed but cannot be replaced. Circular chromosomes don't have ends.

9. In *E. coli*, the origin of replication is called:

- A) ARS
- B) oriC
- C) ORC
- D) TER

✓ **Answer: B** — oriC is the single origin in *E. coli*. ARS (Autonomously Replicating Sequence) is used in yeast; ORC is the origin recognition complex in eukaryotes.

10. Semiconservative replication was proven by:

- A) Watson and Crick
- B) Kornberg

- C) Meselson and Stahl
- D) Okazaki

✓ **Answer: C** — Meselson and Stahl (1958) used $^{15}\text{N}/^{14}\text{N}$ density gradient centrifugation to demonstrate semiconservative replication.

Questions 11-20: Prokaryotic Replication

11. DnaA protein functions in:

- A) Unwinding DNA at the replication fork
- B) Synthesizing RNA primers
- C) Initiating replication at oriC
- D) Joining Okazaki fragments

✓ **Answer: C** — DnaA-ATP binds DnaA boxes at oriC and induces melting of the AT-rich region, initiating replication.

12. DnaB helicase moves in which direction on DNA?

- A) $3' \rightarrow 5'$
- B) $5' \rightarrow 3'$
- C) Bidirectionally
- D) Direction depends on strand

✓ **Answer: B** — DnaB is a $5' \rightarrow 3'$ helicase. It encircles the lagging strand template and translocates $5' \rightarrow 3'$ to unwind DNA.

13. The function of DnaC is to:

- A) Synthesize RNA primers
- B) Load DnaB helicase onto DNA
- C) Remove RNA primers
- D) Seal nicks between Okazaki fragments

✓ **Answer: B** — DnaC is a helicase loader that delivers DnaB to single-stranded DNA at the origin. It holds DnaB inactive during loading, then dissociates.

14. The τ (tau) subunit of Pol III is important for:

- A) Proofreading
- B) Dimerizing two Pol III cores (one per strand)
- C) Loading the β clamp
- D) Removing RNA primers

✓ **Answer: B** — τ subunit dimerizes two Pol III core complexes and connects them to DnaB helicase, coordinating leading and lagging strand synthesis.

15. What is the size of Okazaki fragments in *E. coli*?

- A) 100-200 nt
- B) 500-700 nt

- C) 1,000-2,000 nt
- D) 5,000-10,000 nt

✓ **Answer: C** — *E. coli* Okazaki fragments are 1,000-2,000 nt. In eukaryotes, they are 100-200 nt.

16. In *E. coli* mismatch repair, strand discrimination is based on:

- A) Strand breaks at Okazaki fragment junctions
- B) Dam methylation of GATC sequences
- C) PCNA orientation
- D) Histone modification

✓ **Answer: B** — *E. coli* uses Dam methylation. The parental strand is methylated; the new strand is temporarily unmethylated (hemimethylated). MutH nicks the unmethylated strand.

17. The Tus/Ter system in *E. coli* functions to:

- A) Initiate replication
- B) Increase fork speed
- C) Trap replication forks at termination
- D) Remove RNA primers

✓ **Answer: C** — Tus binds Ter sequences and creates a polar fork barrier, trapping converging forks in the termination region.

18. DNA ligase in *E. coli* uses which cofactor?

- A) ATP
- B) GTP
- C) NAD⁺
- D) FAD

✓ **Answer: C** — *E. coli* DNA ligase uses NAD⁺. Eukaryotic DNA ligases use ATP. This is a key biochemical difference.

19. The "trombone model" describes:

- A) How helicase moves along DNA
- B) How the lagging strand template loops to coordinate synthesis with the leading strand
- C) How primase interacts with DnaB
- D) How topoisomerase relieves supercoiling

✓ **Answer: B** — The trombone model describes how the lagging strand template loops back so that both Pol III cores at the fork can physically move in the same direction.

20. DNA gyrase is inhibited by:

- A) Rifampicin
- B) Fluoroquinolones (ciprofloxacin)
- C) Ampicillin
- D) Tetracycline

✓ **Answer: B** — Fluoroquinolones trap the gyrase-DNA covalent complex, causing lethal double-strand breaks. Rifampicin targets RNA polymerase.

Questions 21-30: Eukaryotic Replication

21. The eukaryotic sliding clamp is:

- A) β_2 homodimer
- B) PCNA homotrimer
- C) RFC pentamer
- D) MCM2-7 hexamer

✓ **Answer: B** — PCNA (Proliferating Cell Nuclear Antigen) is a homotrimer forming a ring, similar in function to the bacterial β_2 dimer.

22. Which eukaryotic polymerase synthesizes the leading strand?

- A) Pol α
- B) Pol δ
- C) Pol ϵ
- D) Pol γ

✓ **Answer: C** — Pol ϵ is the primary leading strand polymerase. Pol δ synthesizes the lagging strand. Pol α /primase initiates synthesis.

23. What is the function of geminin?

- A) Activates origins in S phase
- B) Inhibits Cdt1 to prevent re-replication
- C) Loads MCM helicase
- D) Phosphorylates CDK substrates

✓ **Answer: B** — Geminin binds and inhibits Cdt1 in S/G2 phase, preventing new MCM loading and thus preventing re-replication.

24. The CMG helicase consists of:

- A) DnaB + DnaC + SSB
- B) Cdc45 + MCM2-7 + GINS
- C) ORC + Cdc6 + Cdt1
- D) Pol ϵ + PCNA + RFC

✓ **Answer: B** — CMG = Cdc45-MCM-GINS. It is the active replicative helicase in eukaryotes, formed when Cdc45 and GINS bind phosphorylated MCM2-7.

25. PCNA is loaded onto DNA by:

- A) Pol α /primase
- B) RPA
- C) RFC (Replication Factor C)
- D) MCM2-7

✓ **Answer: C** — RFC (clamp loader) uses ATP hydrolysis to open the PCNA ring and load it onto primer-template junctions.

26. RNase H2 in eukaryotes functions to:

- A) Remove DNA mismatches
- B) Remove RNA primers during Okazaki fragment processing
- C) Unwind DNA at the replication fork
- D) Synthesize RNA primers

✓ **Answer: B** — RNase H2 degrades most of the RNA primer, leaving 1-2 ribonucleotides that are then removed by FEN1.

27. Replication licensing occurs in which cell cycle phase?

- A) S phase
- B) G2 phase
- C) M phase
- D) G1 phase

✓ **Answer: D** — Licensing occurs in G1 when CDK activity is low. Pre-RC assembly (ORC → Cdc6 → Cdt1 → MCM) can only occur when CDK is inactive.

28. How many origins of replication does a human cell have approximately?

- A) 1
- B) 100
- C) 30,000-50,000
- D) 1,000,000

✓ **Answer: C** — Human cells have approximately 30,000-50,000 origins. This allows the large genome to be replicated in ~6-8 hours.

29. The size of Okazaki fragments in human cells is approximately:

- A) 1,000-2,000 nt
- B) 500-1,000 nt
- C) 100-200 nt
- D) 50-100 nt

✓ **Answer: C** — Human Okazaki fragments are 100-200 nt, much shorter than E. coli's 1-2 kb fragments (due to nucleosome obstacles).

30. FEN1 functions in:

- A) Unwinding DNA
- B) Removing RNA primers and DNA flaps
- C) Loading PCNA
- D) Synthesizing RNA primers

✓ **Answer: B** — FEN1 (Flap Endonuclease 1) removes 5' flap structures created during Okazaki fragment processing after Pol δ displacement synthesis.

Questions 31-40: Advanced Topics

31. Lynch syndrome is caused by mutations in:

- A) BRCA1/BRCA2
- B) MLH1/MSH2/MSH6/PMS2
- C) TP53
- D) APC

✓ **Answer: B** — Lynch syndrome (HNPCC) is caused by germline mutations in mismatch repair genes (MLH1, MSH2, MSH6, PMS2), leading to microsatellite instability.

32. Telomerase is:

- A) A DNA-dependent DNA polymerase
- B) An RNA-dependent DNA polymerase (reverse transcriptase)
- C) An RNA-dependent RNA polymerase
- D) A DNA-dependent RNA polymerase

✓ **Answer: B** — Telomerase (TERT) is a reverse transcriptase that uses its RNA component (TERC) as a template to synthesize telomeric DNA.

33. The TERT promoter mutations found in cancer:

- A) Inactivate telomerase
- B) Create new transcription factor binding sites → increase TERT expression
- C) Change the telomere repeat sequence
- D) Delete the entire TERT gene

✓ **Answer: B** — C228T and C250T mutations create new ETS/GABP binding sites, increasing TERT transcription and telomerase activity.

34. ALT (Alternative Lengthening of Telomeres) uses:

- A) Telomerase
- B) Homologous recombination
- C) Translesion synthesis
- D) Non-homologous end joining

✓ **Answer: B** — ALT uses homologous recombination between telomeric sequences to maintain telomeres without telomerase. It results in highly heterogeneous telomere lengths.

35. The ATR-Chk1 checkpoint is activated by:

- A) Double-strand breaks
- B) Excessive single-stranded DNA at stalled replication forks
- C) Unattached kinetochores
- D) Telomere shortening

✓ **Answer: B** — ATR is activated by RPA-coated ssDNA, which accumulates at stalled replication forks. ATRIP binds RPA-ssDNA and recruits ATR.

36. Translesion synthesis differs from template switching in that:

- A) TLS is error-free; template switching is error-prone
- B) TLS uses specialized polymerases; template switching uses the sister chromatid
- C) TLS requires PCNA; template switching does not
- D) TLS only occurs in bacteria

✓ **Answer: B** — Translesion synthesis uses low-fidelity specialized polymerases (Pol η , ι , κ) to bypass lesions (error-prone). Template switching uses the intact sister chromatid as template (error-free).

37. Common fragile sites in cancer are:

- A) Regions where tumor suppressors are located
- B) Large, gene-rich regions prone to breakage under replication stress
- C) Regions where telomeres are located
- D) Origins of replication

✓ **Answer: B** — Common fragile sites are large genomic regions (>1 Mb genes) that are difficult to replicate. They show gaps/breaks under replication stress and are frequent breakpoints in cancer.

38. Oncogene-induced replication stress is caused by:

- A) Telomere shortening
- B) Excessive origin firing, transcription-replication conflicts, and dNTP imbalance
- C) Loss of p53
- D) Telomerase activation

✓ **Answer: B** — Oncogenes (Ras, Myc, Cyclin E) cause aberrant origin firing, increase transcription, and create dNTP imbalances, all contributing to replication stress.

39. BRCA1/BRCA2 mutations increase cancer risk because they impair:

- A) Mismatch repair
- B) Base excision repair
- C) Homologous recombination
- D) Nucleotide excision repair

✓ **Answer: C** — BRCA1 and BRCA2 are essential for homologous recombination. Loss of HR → genome instability → cancer predisposition (breast, ovarian).

40. PARP inhibitors exploit synthetic lethality in:

- A) All cancer cells
- B) BRCA1/BRCA2-mutant cancer cells
- C) Telomerase-positive cancer cells
- D) MSI-High cancer cells

✓ **Answer: B** — PARP inhibitors prevent SSB repair → SSBs converted to DSBs during replication → in BRCA-deficient cells (no HR), DSBs cannot be repaired → cell death. Normal cells survive because HR is intact.

Total: 40 MCQs covering all major topics

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DNA Replication — Short Answer Questions

25 Questions (2-5 marks each)

Question 1 (3 marks)

Explain why DNA replication is described as "semiconservative." What would you expect to see after two rounds of replication in the Meselson-Stahl experiment if replication were conservative instead?

Question 2 (3 marks)

List the three layers of replication fidelity in *E. coli* and explain how each contributes to overall accuracy.

Question 3 (3 marks)

Why does DNA polymerase require a primer? Explain the chemical basis for this requirement.

Question 4 (3 marks)

Compare the structure and function of the β sliding clamp (*E. coli*) and PCNA (eukaryotes).

Question 5 (4 marks)

Describe the role of DnaA protein in *E. coli* replication initiation. What happens if DnaA cannot bind ATP?

Question 6 (3 marks)

Explain why the lagging strand is synthesized discontinuously. What would happen if both strands could be synthesized continuously?

Question 7 (4 marks)

Describe the function of DNA polymerase I in *E. coli*. Why is it unique among *E. coli* polymerases?

Question 8 (3 marks)

What is the role of DNA gyrase in replication? How do fluoroquinolone antibiotics exploit this?

Question 9 (4 marks)

Explain the concept of replication licensing. Why is it important for genome stability?

Question 10 (3 marks)

Describe the differences between Pol ϵ and Pol δ in terms of their roles during eukaryotic replication.

Question 11 (4 marks)

Explain the end-replication problem. Why does it affect linear chromosomes but not circular chromosomes?

Question 12 (3 marks)

What is the role of telomerase in maintaining telomere length? Name its two essential components.

Question 13 (4 marks)

Explain how the ATR-Chk1 checkpoint pathway responds to replication stress.

Question 14 (3 marks)

Compare the mismatch repair systems in *E. coli* and human cells. What is the key difference in strand discrimination?

Question 15 (4 marks)

Describe how Okazaki fragments are processed in eukaryotes. Name the enzymes involved at each step.

Question 16 (3 marks)

What is chromatin remodeling during replication? Why is it necessary in eukaryotes but not prokaryotes?

Question 17 (3 marks)

Explain the relationship between CDK activity and replication licensing. Why are these mutually exclusive?

Question 18 (4 marks)

A researcher discovers a new protein that, when mutated, causes cells to accumulate reversed replication forks. What protein might this be, and what is its normal function?

Question 19 (3 marks)

Describe the mechanism by which translesion synthesis bypasses DNA lesions.

Question 20 (4 marks)

Explain how oncogene activation leads to replication stress. Give two specific examples of oncogenes and their mechanisms.

Question 21 (3 marks)

What is the role of the Shelterin complex at telomeres? Name three of its components.

Question 22 (3 marks)

Compare the replication speeds of *E. coli* and human cells. Explain why they differ.

Question 23 (4 marks)

Describe the complete pathway from origin recognition to fork activation in eukaryotic cells.

Question 24 (3 marks)

How does the Fanconi anemia pathway relate to DNA replication? What happens when it is defective?

Question 25 (3 marks)

Explain the concept of synthetic lethality and how it is applied in cancer therapy using PARP inhibitors.

Suggested Marking Scheme

3-mark questions: 1 mark for each key point **4-mark questions:** 1 mark for each key point + clarity marks

Key Points for Sample Answers

Q1 (Semiconservative):

- Each daughter = 1 old + 1 new strand (1)
- Conservative: would show heavy + light bands (not hybrid) (1)
- Dispersive: would show intermediate density bands (1)

Q9 (Licensing):

- ORC/Cdc6/Cdt1/MCM assembly (1)
- G1 phase (low CDK) (1)
- Once per cycle (1)
- Prevents re-replication (1)

Q11 (End-replication):

- RNA primer removal at 5' end (1)
- No 3'-OH available to fill gap (1)
- Progressive shortening (1)
- Circular = no ends (1)

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DNA Replication — Long Answer / Essay Questions

15 Questions (8-12 marks each)

Question 1 (10 marks)

Describe the complete mechanism of DNA replication initiation in *E. coli*, from DnaA binding to the formation of two functional replication forks. Include the roles of all major proteins involved.

Expected content: DnaA-ATP, DnaA boxes, AT-rich melting, DnaC/DnaB loading, bidirectional fork formation, helicase activation

Question 2 (10 marks)

Compare and contrast the replication mechanisms of *E. coli* and human cells. Include at least eight specific comparisons covering initiation, elongation, termination, and regulation.

Expected content: Origins, helicases, polymerases, sliding clamps, Okazaki fragment size, telomeres, licensing, chromatin, speed

Question 3 (12 marks)

Explain the concept of replication licensing in eukaryotic cells. Describe the molecular events that occur during G₁ licensing and S-phase activation. Why is this system essential for maintaining genome stability? What happens when it fails?

Expected content: Pre-RC assembly, ORC/Cdc6/Cdt1/MCM, CDK regulation, geminin, once-per-cycle firing, re-replication consequences

Question 4 (10 marks)

Describe the three layers of replication fidelity in *E. coli*. Explain the molecular mechanism at each level and calculate the overall error rate. What are the clinical consequences of defects in each layer?

Expected content: Base selection, proofreading, MMR, error rates, Lynch syndrome, cancer predisposition

Question 5 (10 marks)

Explain the end-replication problem in detail. How do telomeres and telomerase address this problem? Describe the structure and mechanism of telomerase and discuss its role in aging and cancer.

Expected content: Linear chromosome problem, primer removal, progressive shortening, telomerase structure (TERT/TERC), extension mechanism, aging, cancer, ALT

Question 6 (10 marks)

Describe the structure and function of the eukaryotic replisome during S phase. Include the roles of CMG helicase, RPA, Pol α /primase, Pol ϵ , Pol δ , PCNA, and RFC. How are these components coordinated?

Expected content: CMG activation, polymerase switching, leading/lagging strand synthesis, Okazaki fragment processing

Question 7 (12 marks)

Explain the ATR-CHK1 checkpoint pathway in response to replication stress. What types of stress activate this pathway? What are the downstream effects? How does this pathway relate to cancer development and therapy?

Expected content: ssDNA/RPA, ATRIP recruitment, CHK1 activation, fork stabilization, origin suppression, cancer relevance, therapeutic targeting

Question 8 (10 marks)

Discuss the role of mismatch repair in maintaining genome stability. Compare bacterial (MutHLS) and eukaryotic (MutS α /MutL α) mismatch repair systems. Explain the molecular basis of Lynch syndrome and describe how microsatellite instability is diagnosed.

Expected content: MMR mechanism, strand discrimination (methylation vs nicks), Lynch syndrome, MSI testing, cancer predisposition

Question 9 (10 marks)

Explain how chromatin is disassembled and reassembled during eukaryotic DNA replication. What are the roles of FACT, CAF-1, and ASF1? How are parental histone marks inherited by daughter chromosomes?

Expected content: Nucleosome disassembly, histone chaperones, recycling, reassembly, epigenetic inheritance, replication timing

Question 10 (12 marks)

Describe the molecular mechanisms by which replication stress contributes to genome instability and cancer. Include discussions of fork stalling, fork reversal, replication-transcription conflicts, and common fragile sites.

Expected content: Causes of stress, fork dynamics, R-loops, fragile sites, oncogene-induced stress, genome instability cascade

Question 11 (10 marks)

Explain the mechanism of translesion synthesis and template switching. How does the cell decide between these two pathways? What are the consequences of each for genome stability? Give specific examples of TLS polymerases and their functions.

Expected content: PCNA ubiquitylation, Pol η / ι / κ , error-prone vs error-free, lesion bypass, mutagenesis

Question 12 (10 marks)

Compare and contrast the Tus/Ter termination system in *E. coli* with the fork convergence termination mechanism in eukaryotes. What happens to the replisome after termination? How are daughter chromosomes separated?

Expected content: Tus/Ter polar trap, CMG disassembly (p97), decatenation (Topo II), chromosome segregation

Question 13 (8 marks)

Explain how different antibiotics and anticancer drugs target DNA replication. Include at least four specific drug targets and describe the mechanism of action for each.

Expected content: Fluoroquinolones (gyrase), PARP inhibitors (synthetic lethality), nucleoside analogs (chain termination), ATR/CHK1 inhibitors

Question 14 (10 marks)

Discuss the evolution of DNA replication machinery from prokaryotes to eukaryotes. What features are conserved? What features are unique to each system? How does the archaeal replication system relate to both?

Expected content: Conservation of sliding clamp, clamp loader, helicase family, polymerases; differences in origin structure, licensing, chromatin

Question 15 (12 marks)

A researcher discovers a new genetic disease characterized by premature aging, bone marrow failure, and increased cancer risk. Patient cells show extremely short telomeres and mutations in a protein that interacts with telomerase. Design a comprehensive experimental plan to characterize this disease, including molecular, cellular, and clinical approaches.

Expected content: Telomere length measurement, telomerase activity assays, gene identification, functional studies, mouse models, potential therapies

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DNA Replication — Viva Voce Questions

30 Questions for Oral Examination

Basic Concepts (1-8)

1. What is DNA replication and why is it essential for life?

- *Expected answer:* Semiconservative copying of genome before cell division; ensures each daughter cell gets complete genetic information

2. What are the three models of DNA replication? Which was proven correct?

- *Expected answer:* Conservative, semiconservative, dispersive. Semiconservative proven by Meselson-Stahl (1958)

3. Can DNA polymerase synthesize DNA in the 3'→5' direction? Why or why not?

- *Expected answer:* No. Chemical mechanism requires 3'-OH for nucleophilic attack on incoming dNTP. Universal rule in all organisms.

4. What is the difference between a leading strand and a lagging strand?

- *Expected answer:* Both synthesized 5'→3'. Leading: continuous toward fork. Lagging: discontinuous (Okazaki fragments) away from fork.

5. Why do we need RNA primers for DNA replication?

- *Expected answer:* DNA polymerase cannot initiate synthesis de novo; needs free 3'-OH. Primase provides this with RNA primer.

6. What is the sliding clamp and why is it important?

- *Expected answer:* Ring-shaped protein (β_2 in bacteria, PCNA in eukaryotes) that tethers polymerase to template. Increases processivity from ~20 to >50,000 nt.

7. What is proofreading and how does it improve replication fidelity?

- *Expected answer:* 3'→5' exonuclease activity of polymerase removes misincorporated nucleotides. Improves fidelity 100-fold (10^{-5} to 10^{-7}).

8. What is the overall error rate of DNA replication after all correction mechanisms?

- *Expected answer:* $\sim 10^{-9}$ to 10^{-10} per nucleotide per cell division. Three layers: selection (10^{-5}), proofreading (10^{-7}), MMR (10^{-9} – 10^{-10})

Prokaryotic Replication (9-15)

9. Describe the structure and function of oriC.

- *Expected answer:* 245 bp origin; DnaA boxes (9-mers) for DnaA binding; AT-rich 13-mers for melting. Single origin in *E. coli*.

10. What is the role of DnaA in replication initiation?

- *Expected answer:* Initiator protein. DnaA-ATP binds DnaA boxes cooperatively, induces positive supercoiling, melts AT-rich region.

11. Explain how DnaB helicase is loaded onto DNA.

- *Expected answer:* DnaC holds DnaB inactive → delivers to melted origin → DnaB encircles ssDNA → DnaC dissociates (ATP-dependent)

12. What is unique about DNA Pol I compared to other *E. coli* polymerases?

- *Expected answer:* Has 5'→3' exonuclease activity for RNA primer removal. Also has polymerase and 3'→5' exonuclease (proofreading).

13. Why does DNA ligase in *E. coli* use NAD⁺ instead of ATP?

- *Expected answer:* Biochemical difference from eukaryotes. Same mechanism (adenylation) but different cofactor. Eukaryotic/viral ligases use ATP.

14. How does the Tus/Ter system ensure termination occurs in the correct region?

- *Expected answer:* Tus binds Ter sequences → polar fork barrier → blocks fork in one direction but not other → traps both forks in termination region.

15. What would happen if DNA gyrase were inactivated in *E. coli*?

- *Expected answer:* Positive supercoils accumulate ahead of fork → fork stalling → DNA breakage → cell death. Gyrase is essential.

Eukaryotic Replication (16-22)

16. What is replication licensing and when does it occur?

- *Expected answer:* Assembly of pre-RC (ORC→Cdc6→Cdt1→MCM) in G₁ when CDK low. Ensures each origin fires once per cell cycle.

17. Why do eukaryotes need multiple origins of replication?

- *Expected answer:* Large genome (3.2×10^9 bp); single origin would take weeks. Multiple origins (~30,000-50,000) allow replication in 6-8 hours.

18. What is the CMG helicase and how is it activated?

- *Expected answer:* Cdc45-MCM2-7-GINS. Inactive MCM loaded at origins → DDK + CDK phosphorylate → Cdc45 and GINS bind → active helicase.

19. Name the three major eukaryotic replicative polymerases and their functions.

- *Expected answer:* Pol α /primase (initiates), Pol ϵ (leading strand), Pol δ (lagging strand + repair)

20. What is the role of PCNA in mismatch repair?

- *Expected answer:* RFC loads PCNA in oriented manner; MutLa recognizes PCNA orientation → nicks new strand → directs repair to correct strand.

21. How are Okazaki fragments processed in eukaryotes?

- *Expected answer:* RNase H2 removes most RNA → FEN1 removes remaining RNA + flaps → Pol δ fills gaps → DNA Ligase I seals nicks

22. What is the difference between early and late replication in S phase?

- *Expected answer:* Euchromatin (active genes) replicates early; heterochromatin replicates late. Correlates with gene expression and epigenetic state.

Telomeres and Checkpoints (23-27)

23. Explain the end-replication problem in one sentence.

- *Expected answer:* Linear chromosomes cannot fully replicate at 5' ends because RNA primers at the terminus are removed but cannot be replaced.

24. What are the two essential components of telomerase?

- *Expected answer:* TERT (telomerase reverse transcriptase, protein) and TERC (telomerase RNA component, contains template)

25. Why do most somatic cells not express telomerase?

- *Expected answer:* Telomere shortening acts as a biological clock limiting cell divisions (tumor suppression). Stem cells and germ cells maintain telomerase.

26. How is the ATR-Chk1 checkpoint activated?

- *Expected answer:* Stalled fork → ssDNA accumulates → RPA binds → ATRIP recruits ATR → ATR phosphorylates Chk1 → fork stabilization, origin suppression

27. What is the clinical significance of the TERT C228T promoter mutation?

- *Expected answer:* Creates ETS binding site → increased TERT transcription → telomerase reactivation → immortalization → found in many cancers

Applications and Disease (28-30)

28. How do fluoroquinolone antibiotics target bacterial replication?

- *Expected answer:* Trap gyrase-DNA complex → stabilize DSBs → fork collapse → cell death. Selective for bacterial gyrase over human topoisomerases.

29. Explain how PARP inhibitors work in BRCA-mutant cancers.

- *Expected answer:* Synthetic lethality. BRCA-mutant cells lack HR; PARP inhibition prevents SSB repair → SSBs become DSBs → cannot repair → cell death.

30. A patient has microsatellite instability in their colorectal cancer. What gene mutations would you suspect?

- *Expected answer:* Mismatch repair genes (MLH1, MSH2, MSH6, PMS2). Likely Lynch syndrome (HNPCC). Recommend genetic testing and family screening.

Bonus Questions (31-33)

31. What would happen if all DnaA in *E. coli* were replaced with a mutant that cannot hydrolyze ATP?

- *Expected answer:* DnaA-ATP would accumulate → continuous origin firing → re-replication → gene amplification → genome instability → likely lethal

32. How does the cell prevent the lagging strand Pol III from continuously falling off the template?

- *Expected answer:* β sliding clamp increases processivity; trombone model coordinates with leading strand; τ subunit links both polymerases

33. A research lab discovers a protein that binds specifically to replication forks and promotes fork reversal. What would be the phenotype if this protein were knocked out?

- *Expected answer:* Reduced fork reversal → increased fork collapse → more DSBs → genome instability → sensitivity to replication stress

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DNA Replication — Case Studies

10 Case Studies requiring critical reasoning

Case Study 1: The Young Cancer Patient

Clinical Presentation: A 28-year-old woman presents with early-onset colorectal cancer. Her family history includes:

- Maternal aunt: ovarian cancer (age 35)
- Maternal grandmother: endometrial cancer (age 50)
- Brother: gastric cancer (age 30)

Laboratory Findings:

- Tumor biopsy: MSI-High
- MLH1 immunohistochemistry: negative in tumor
- MLH1 promoter methylation: negative
- Germline sequencing: MLH1 c.676C>T (p.Arg226*)

Questions:

1. What is the most likely diagnosis? Explain the molecular basis.
2. Why is MLH1 promoter methylation negative? What is the significance?
3. What is the recurrence risk for the patient's children?
4. What surveillance recommendations would you make for the patient's siblings?
5. What immunotherapy might be effective and why?

Case Study 2: The Premature Aging Syndrome

Clinical Presentation: A 15-year-old boy presents with:

- Short stature, microcephaly
- Bone marrow failure (pancytopenia)
- Pulmonary fibrosis
- Very short telomeres (<3 kb; normal: 8-12 kb)

Genetic Testing:

- TERT: c.2282G>A (p.Gly761Glu) — heterozygous
- TERC: normal
- DKC1: normal

Questions:

1. What is the likely diagnosis?

2. How does the TERT mutation affect telomerase function?
3. Why does this patient have bone marrow failure?
4. Would bone marrow transplantation be effective? What are the considerations?
5. How does this condition relate to normal aging?

Case Study 3: The Replication Mutant

Research Scenario: A researcher isolates an *E. coli* mutant with the following phenotype:

- Extremely high mutation rate (100-fold increase)
- Okazaki fragments are normal size (1-2 kb)
- Leading strand synthesis appears normal
- Lagging strand synthesis shows gaps between Okazaki fragments
- No growth on minimal media

Questions:

1. Which protein is likely mutated? Explain your reasoning.
2. Why are Okazaki fragments normal size but gaps persist?
3. Why is the mutation rate increased?
4. Design an experiment to confirm your hypothesis.
5. What would happen if you introduced a wild-type copy of this gene on a plasmid?

Case Study 4: The Cancer Drug Response

Clinical Scenario: A pharmaceutical company tests Drug X on cancer cell lines:

Cell Line	BRCA1 Status	Drug X IC50	γH2AX Foci
Cell A	Wild type	10 μM	High
Cell B	Mutant	50 μM	Moderate
Cell C	Wild type	8 μM	High
Cell D	Mutant	45 μM	Moderate

Questions:

1. What is the likely mechanism of Drug X?
2. Why are BRCA1-mutant cells more resistant?
3. Would you develop Drug X as a monotherapy? Why or why not?
4. Design a combination therapy strategy using Drug X.
5. What biomarker would you use to select patients?

Case Study 5: The Telomerase Mystery

Research Scenario: A researcher studies two cancer cell lines:

- **Line 1:** Telomerase-positive, stable telomeres (8 kb)

- **Line 2:** Telomerase-negative, highly heterogeneous telomeres (2-25 kb), APBs positive, C-circles positive

Questions:

1. What telomere maintenance mechanism does Line 2 use?
2. What molecular markers confirm this?
3. How does ALT differ from telomerase-mediated maintenance?
4. Would telomerase inhibitors be effective against Line 2?
5. What therapeutic approach might work for ALT-positive cancers?

Case Study 6: The Replication Fork Analysis

Experimental Data: A researcher performs DNA fiber analysis on cells treated with Drug Y:

Condition	IdU track (μm)	CldU track (μm)	Pattern
Control	11.0 ± 1.5	11.5 ± 1.2	Symmetric
Drug Y (low)	10.8 ± 1.4	5.2 ± 1.8	Asymmetric
Drug Y (high)	10.5 ± 1.3	2.1 ± 0.9	Asymmetric
Drug Y + Rad51 inhibitor	10.2 ± 1.5	0.5 ± 0.3	Collapsed

Questions:

1. What does the asymmetric pattern tell you about Drug Y's mechanism?
2. Why does adding a Rad51 inhibitor cause fork collapse?
3. What is the relationship between fork reversal and fork collapse?
4. Design an experiment to determine the specific target of Drug Y.
5. How would you predict Drug Y would work in BRCA-deficient cells?

Case Study 7: The Okazaki Fragment Puzzle

Experimental Data: A researcher analyzes Okazaki fragments from patient cells:

Sample	Fragment Size	Primer Removal	Ligation	MMR Status
Normal	150 ± 30 nt	Normal	Normal	Normal
Patient 1	250 ± 50 nt	Defective	Normal	Normal
Patient 2	150 ± 30 nt	Normal	Defective	Normal
Patient 3	150 ± 30 nt	Normal	Normal	Defective

Questions:

1. What gene is likely mutated in Patient 1?
2. What gene is likely mutated in Patient 2?
3. What gene is likely mutated in Patient 3?
4. For each patient, describe the predicted clinical phenotype.
5. How would you confirm each diagnosis?

Case Study 8: The Origin of Replication Problem

Research Scenario: A researcher deletes a specific origin (Ori-1) from a yeast chromosome:

- Wild type: Origin fires in early S phase
- Δ Ori-1: Region replicates in late S phase from adjacent origins
- No growth defect at normal temperature
- Growth defect at 37°C
- Increased chromosome breaks at the deleted origin region

Questions:

1. Why does deletion of a single origin not prevent replication of that region?
2. Why is there a growth defect at high temperature?
3. Why are chromosome breaks increased at the deletion region?
4. What does this tell you about the relationship between origin usage and genome stability?
5. How might this relate to common fragile sites in human cells?

Case Study 9: The DNA Polymerase Switch

Experimental Data: A researcher creates temperature-sensitive mutations in yeast Pol ϵ and Pol δ :

Mutation	Leading Strand	Lagging Strand	Viability
pol2-ts (Pol ϵ)	Defective	Normal	Lethal at 37°C
pol3-ts (Pol δ)	Normal	Defective	Lethal at 37°C
pol2-K620A (no Pol ϵ -Pol δ interaction)	Slow	Slow	Slow growth

Questions:

1. Why is loss of Pol ϵ lethal while loss of Pol δ might not be (in some contexts)?
2. What does the double-slow phenotype of the interaction mutant tell you?
3. How do cells compensate when one polymerase is partially impaired?
4. What would you predict for a pol2-ts;pol3-ts double mutant?
5. How does this experiment support the leading/lagging strand polymerase assignment?

Case Study 10: The Cancer Genomics Case

Data Analysis: A researcher analyzes whole-genome sequencing data from a tumor:

Feature	Tumor	Normal
SNV burden	15,000	5,000
MSI status	High	Stable
LOH at MMR loci	MLH1 deleted	MLH1 intact
TERT promoter	C228T mutated	Wild type

Feature	Tumor	Normal
Telomere length	12 kb	8 kb
Aneuploidy	Moderate	None

Questions:

1. What explains the high SNV burden?
2. What is the relationship between MLH1 deletion and MSI?
3. Why are telomeres longer in the tumor despite TERT promoter mutation?
4. What is the significance of aneuploidy?
5. Design a treatment strategy for this patient based on the molecular profile.

Instructor Answer Key Summary

Case 1

1. **Lynch syndrome** — germline MLH1 mutation (pathogenic frameshift)
2. **Methylation negative** = hereditary (not sporadic); sporadic cases show MLH1 promoter methylation
3. **50%** risk for children (autosomal dominant)
4. **Annual colonoscopy from age 20-25**; endometrial screening
5. **Anti-PD-1** (pembrolizumab) — MSI-H = high neoantigen load

Case 2

1. **Dyskeratosis congenita** (TERT mutation)
2. **Gly761Glu** affects TERT catalytic activity → reduced telomere extension
3. **Bone marrow** has high cell turnover → most sensitive to telomere shortening
4. **BMT may help** but conditioning regimen must be reduced (treosulfan-based)
5. **Links to normal aging** — telomere shortening is part of aging process

Case 3

1. **Ligase mutant** — gaps between Okazaki fragments (normal fragments but not joined)
2. **Okazaki fragments synthesized normally** but ligase cannot seal nicks
3. **Gaps in DNA** → chromosome breaks during next replication → genome instability
4. **Complementation test** — introduce wild-type ligase gene on plasmid
5. **Should rescue** if plasmid expresses functional ligase

Case 4

1. **PARP inhibitor** — causes DSBs during replication
2. **BRCA1-mutant cells already have HR defect** → PARP inhibition adds less toxicity (but actually more complex — BRCA1-WT cells show more sensitivity because Drug X specifically targets fork restart pathway)
3. **Not as monotherapy** — resistance develops; use in combination
4. **Combine with platinum chemotherapy** or immune checkpoint inhibitor

5. **γH2AX foci, BRCA1/2 status** as biomarkers

Case 5

1. **ALT** (Alternative Lengthening of Telomeres)
2. **APBs** (ALT-associated PML bodies), **C-circles**, heterogeneous telomere length
3. **Homologous recombination** between telomeric sequences (vs. reverse transcriptase)
4. **No** — telomerase inhibitors won't work on telomerase-negative cells
5. **Target recombination** (ATR inhibitors, PARP inhibitors)

Case 6

1. **Fork slowing** — CldU tracks shortened → drug inhibits fork progression
2. **Rad51 needed for fork restart** → without it, stalled forks collapse
3. **Fork reversal is protective** → prevents collapse; loss of reversal = collapse
4. **Genetic screen** — identify genes that rescue Drug Y sensitivity
5. **More sensitive** — BRCA-deficient cells already have fork restart defects

Case 7

1. **Patient 1:** FEN1 or RNase H (primer processing defect)
2. **Patient 2:** DNA Ligase I (ligation defect)
3. **Patient 3:** MSH2/MLH1 (mismatch repair defect)
4. **Patient 1:** Potential cancer predisposition (FEN1 mutations linked to cancer)
5. **Patient 2:** LIG1 syndrome-like phenotype
6. **Patient 3:** Lynch syndrome-like phenotype

Case 8

1. **Adjacent origins compensate** — origins can fire from nearby regions
2. **Higher temperature = slower fork progression** → origin deletion becomes critical
3. **Longer distance between origins** → fork stalling at difficult-to-replicate regions
4. **Origin usage correlates with genome stability** — loss increases fragility
5. **Common fragile sites may correspond to regions with few origins**

Case 9

1. **Pol ε essential for leading strand** — no compensation possible
2. **Physical interaction coordinates polymerase switching** at the fork
3. **Remaining polymerase can partially compensate** (with reduced efficiency)
4. **Likely lethal** — loss of both leading and lagging strand polymerases
5. **Confirms Pol ε = leading, Pol δ = lagging** assignments

Case 10

1. **MLH1 loss** → defective MMR → cannot repair replication errors → SNV accumulation
2. **LOH (loss of heterozygosity)** — second allele deleted → complete MMR loss

3. **TERT promoter mutation** → telomerase reactivation → telomere maintenance
4. **Aneuploidy** → chromosome instability from defective checkpoints
5. **Immunotherapy** (anti-PD-1 for MSI-H) + **PARP inhibitor** if HR-deficient